LAB 05

Experimental Validation of Kirchhoff's Voltage Law(KVL) & Kirchhoff's Current Law (KCL) in Electrical Circuits

OBJECTIVE:

Verify Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) using mesh and nodal analysis of the given circuit

EQUIPMENTS REQUIRED:

Sr. no	Name	Range	Quantity
1	Resistors		
2	DC variable voltage supply		
3	Digital Multimeter		
4	Others		

THEORY:

1. **KIRCHHOFF'S VOLTAGE LAW (KVL)**

   Kirchhoff's Voltage Law **"**states that the algebraic sum of all the voltages around any closed path (loop or mesh) is zero".

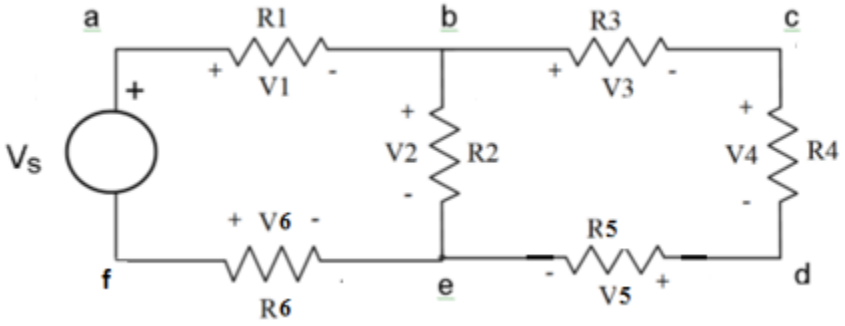


FIGURE 1.

Applying Kirchhoff's voltage law to the first and the second loops in the circuit shown in Figure 1 yields the equation below respectively

$$V_1 + V_2 - V_6 \ -V_S = 0 \qquad \text{(LOOP "a b e f a")} \qquad \text{(1a)}$$

$$V_3 + V_4 + V_5 - V_2 = 0 \qquad \text{(LOOP "b c d e b")} \qquad (1b)$$

BREAD BOARD LAYOUT

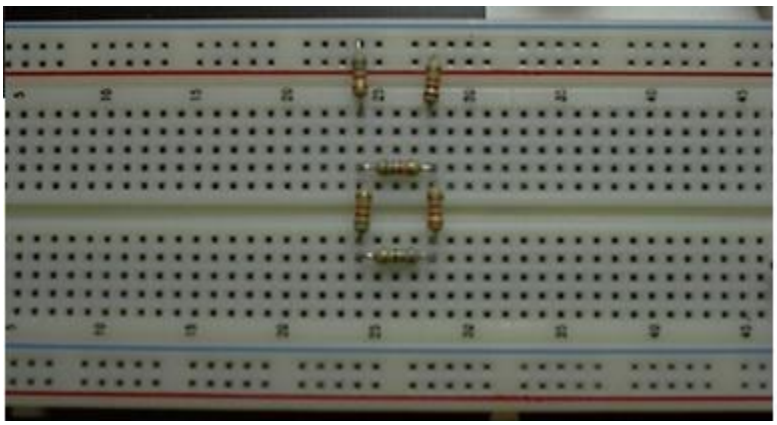


FIGURE 2.

PROCEDURE: KIRCHOFF'S VOLTAGE LAW (KVL)

1. Construct the circuit shown in Figure 1 using the values below:

R1 = 1 KΩ R2 = 2.2 KΩ

R3 = 1.8 KΩ R4 = 1 KΩ

R5 = 1.5 KΩ R6 = 2.7 KΩ

2. Set the Variable Power Supply (Vs) to 5 Volts.

3. Accurately measure all voltages in the circuit using the Digital Multi-Meter (DMM).

4. Record the measurements in the observation table containing the measured voltage values as shown below.

5. Verify the results theoretically and record the value in the observation in the column for calculated value.

6. Verify KVL for the loops in the circuit using equations '1a' and '1b'.

OBSERVATION TABLE:

Element	Specified value	Element Voltage	Element voltage				LOOP VOLTAGE "**a b e f a**" $V_1 + V_2 - V_6 - V_S = 0$
			Nodes		Measured value (V)	Calculated value of voltage (V)	Calculated:
			+	–			

R_1		V_1	a	b			Measured:
R_2		V_2	b	e			
R_3		V_3	b	c			LOOP VOLTAGE "**b c d e b**" $V_3 + V_4 + V_5 - V_2 = 0$
R_4		V_4	c	d			
R_5		V_5	d	e			Calculated:
R_6		V_6	e	f			
Vs		V_s	a	f			Measured:

1. **KIRCHHOFF'S CURRENT LAW (KCL):**

KIRCHHOFF'S CURRENT LAW "states that the algebraic sum of all the currents at any node is zero". Applying Kirchhoff's current law to the nodes in the circuit shown in Figure 3 yields the following equations;

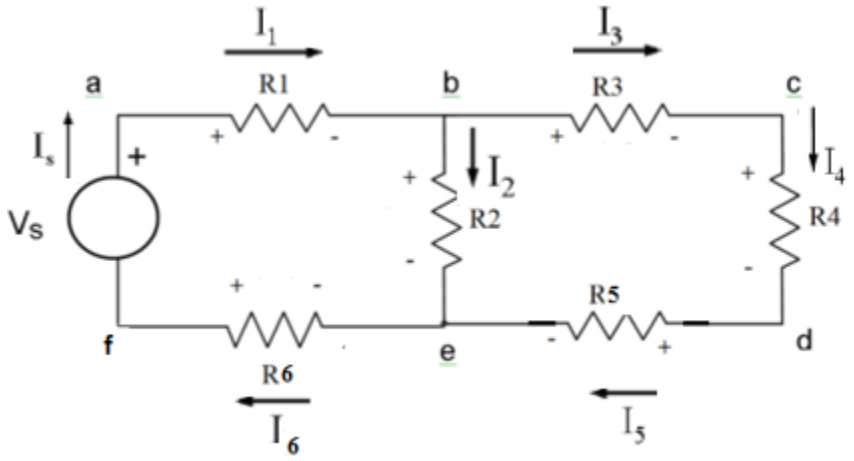


FIGURE 3.

Node a: $-I_s + I_1 = 0$ **(2a)**

Node b: $-I_1 + I_2 + I_3 = 0$ **(2b)**

Node c: $-I_3 + I_4 = 0$ **(2c)**

Node d: $-I_4 + I_5 = 0$ **(2d)**

Node e: $-I_5 - I_2 + I_6 = 0$ **(2e)**

Node f: $+I_s - I_6 = 0$ **(2f)**

PROCEDURE: KIRCHOFF'S Current LAW (KCL)

1. Construct the circuit shown in Figure 3 using the values below:

   R1 = 1 KΩ R2 = 2.2 KΩ

   R3 = 1.8 KΩ R4 = 1 KΩ

   R5 = 1.2 KΩ R6 = 2.7 KΩ

2. Set the Variable Power Supply (Vs) to 5 Volts.

3. Accurately measure all current in each branch using the Digital Multi-Meter (DMM) as shown in figure 4.

4. Record the measurements in the observation table containing the total current into and total current out of respectively.

5. Verify the results theoretically and experimentally and record the value in the observation in the column for calculated and measured value.

6. Verify KCL for the nodes in the circuit using equations '2a' through '2f'.

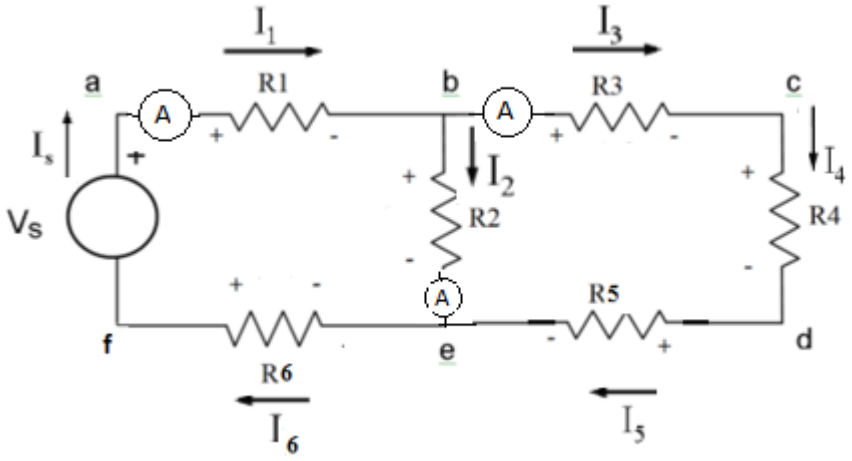


FIGURE 4.

OBSERVATION TABLE:

Node	Total current into (I_{in}) (A)		Total current out of (I_{out}) (A)		KCL ($I_{in} - I_{out}$) (A)	
	Calculated	Measured	Calculated	Measured	Calculated	Measured

a						
b						
c						
d						
e						
f						